Course: BM20A4703 Partial Differential Equations with Applications.
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Worksheet 3

Exercise 2.4

(i) [10] Using the method of characteristics and Rankine-Hugoniot condition, solve explicitly for all $t > 0$ the initial value problem for Burgers' equation

$$u_t + \left(\frac{1}{2}u^2\right)_x = 0, \quad u_0(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0. \end{cases}$$

Hint: The characteristic method does not provide any solution in the strip $-t < x < t$. To fill this strip, try to solve the equation with $u(x,0) = u_0^\epsilon(x)$, where u_0^ϵ is smooth and $u_0^\epsilon \to u_0$ when $\epsilon \to 0$ pointwise. (You may think about some u_0^ϵ that "connects" two discontinuous parts of u_0 by a continuous line).

(ii) [10] Now find a solution for all $t > 0$ when the initial data is

$$u_0(x) = \begin{cases} 1 & \text{if } x < 0 \\ -1 & \text{if } x > 0. \end{cases}$$

Hint: The method of characteristics fails as characteristics crossed immediately. How to find one? Does u_0 satisfy the jump condition? If yes, it is one.

(iii) [10] If we multiply Burgers' equation by u, we formally find that u satisfies

$$\left(\frac{1}{2}u^2\right)_t + \left(\frac{1}{3}u^3\right)_x = 0. \tag{1}$$

Are the solutions to the Burgers equation you found in parts (1) and (2) weak solutions to (1)? If not, find a weak solution to (1).
Hint: What is the Rankine-Hugoniot condition for the PDE (1)? You may mimic the proof of the Rankine-Hugoniot condition in the lecture notes to find it out. See if the condition is satisfied.

Exercise 2.5

[10] Consider the Burgers' equation with initial data u_0,

$$u_t + \left(\frac{1}{2}u^2\right)_x = 0, \quad u_0(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x > 0. \end{cases} \tag{2}$$

Prove that for any $\alpha \in [0,1]$, each u_α provides a weak solution for (2) where

$$u_\alpha(x) = \begin{cases} -1 & \text{if } x < -t \\ x/t & \text{if } -t < x \le -\alpha t \\ -\alpha & \text{if } -\alpha t < x \le 0 \\ \alpha & \text{if } 0 < x \le \alpha t \\ x/t & \text{if } \alpha t < x \le t \\ 1 & \text{if } t < x. \end{cases} \tag{3}$$

Hint: Verify that u_α satifies Rankine-Hugoniot condition at each discontinuous curve.

Exercise 2.6

(i) [10] Assume that the entropy solution to the Burgers equation

$$u_t + \left(\frac{1}{2}u^2\right)_x = 0, \quad u(x,0) = u_0(x),$$

is smooth and compactly supported. Prove that

$$\|u(\cdot,t)\|_{L^2(\mathbb{R})} \le \|u_0\|_{L^2(\mathbb{R})}.$$

Hint: Multiply the equation with u.

(ii) [10] If v is the entropy solution to the Burgers equation corresponding to the initial data v_0. Does the following estimate hold

$$\|u(\cdot,t) - v(\cdot,t)\|_{L^2(\mathbb{R})} \le \|u_0 - v_0\|_{L^2(\mathbb{R})}?$$

Hint: No. Try to find some counterexamples (thinking along the lines of a Riemann problem, i.e., build some u_0, v_0 of simple two-state step functions that you can work out their solutions easily).

Exercise 2.7

[15] Show that if $u \in C^1$ is an entropy solution to $u_t + f(u)_x = 0,\ u(x,0) = u_0(x)$ where u_0 is an increasing function and f is uniformly convex, then u satisfies

$$\frac{u(x+a,t) - u(x,t)}{a} \le \frac{E}{t}, \quad a > 0, t > 0,$$

for some $E > 0$ independent of x, t and a.
Hint: Mean value theorem may help.

Exercise 2.8

[**Optional. You may get bonus points if you solve this exercise.**]
Let u be a weak solution of the form

$$u(x,t) = \begin{cases} u_l & \text{if } x < st \\ u_r & \text{if } x > st, \end{cases} \quad \left(s = \frac{f(u_l) - f(u_r)}{u_l - u_r}\right),$$

of the conservation law $u_t + f(u)_x = 0$ ($x \in \mathbb{R}, t > 0$), where $u_l \ne u_r$ and f is smooth. Prove that if u is an entropy solution, then it satisfies the Oleinik condition

$$\frac{f(u_r) - f(u)}{u_r - u} \le s \le \frac{f(u_l) - f(u)}{u_l - u},$$

for all u in between u_l and u_r.
Hint: Start with the definition of entropy solutions and use integration by parts.

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